Cut Sheet

Board Design Summary

Style: checkerboard
Dimensions: 12.000" (12") × 18.000" (18") × 1.500" (1 1/2")
Species: Walnut, Maple

[Step Diagram]

Cut List

Part	Qty	Width	Thickness	Length	Species
Walnut blank (mill & rip, th	1en crossc	u1t.1002 × "25(4 mm slices)	2133.8634 5"") ("H 1	5131/6.47"1) 9 "	("HWa1l6nut 2 3 / 3 2 ")
Maple blank (mill & rip, the	1n crossc	u1 1032×6225".4 ("mm slices)	2133.8634 5"") ("H 1	5131/6.47"1) 9 "	("HMa1p6le 2 3 / 3 2 ")

Board Layout (Top View)

[Step Diagram]

Step 1: Mill rough stock

Joint one face and one edge on the jointer, then plane to final thickness plus 1/8" (3 mm). Rest overnight, then plane to final thickness. Take ~1/64" (0.4 mm) passes for the last two passes.

Settings:

Final pass depth: ~1/64" (0.4 mm)

WARNING: Use push blocks on the jointer. Never joint stock shorter than 12" (305 mm) or thinner than 1/2" (12.5 mm).

WARNING: Keep hands clear of the infeed and outfeed. Feed only boards at least 12" (305 mm) long.

[Step Diagram]

Step 2: Rip strips

Rip strips with the jointed edge against the fence. Measure from the fence face to the near side of a tooth set toward the fence.

Settings:

Fence to blade: strip width from cut list

WARNING: Use the riving knife and blade guard on every through cut.

WARNING: Use a push stick for any rip narrower than 6" (152 mm). Never reach over or behind the blade.

[Step Diagram]

Step 3: Glue up

Glue the first edge-grain panel. Alternate clamps above and below the panel and check for flat with a straightedge.

Settings:

Clamp spacing: about 8" (200 mm)

NOTE: Use a waterproof PVA glue rated for indirect food contact. Clamp until squeeze-out turns rubbery, then scrape.

[Step Diagram]

Step 4: Flatten first panel

Plane or sand the panel flat. It is still long grain, so the planer is allowed here if the panel is no wider than 12" (305 mm); a wider panel goes to the router sled or drum sander.

WARNING: Keep hands clear of the infeed and outfeed. Feed only boards at least 12" (305 mm) long.

[Step Diagram]

Step 5: Crosscut slices

Crosscut the panel into slices of the final thickness on a crosscut sled with a stop block.

Settings:
Slice length: final board thickness from cut list

DANGER: Never use the rip fence as a length stop for crosscuts. Use a crosscut sled with a stop block set before the blade, so the offcut is never trapped between blade and fence.

WARNING: Do not cut pieces shorter than 6" (152 mm) free-hand on the table saw. Use a sled with a hold-down.

[Step Diagram]

Step 6: Rotate slices

Turn each slice end grain up.

[Step Diagram]

Step 7: Flip alternate slices

Flip every other slice end for end so the colours alternate into squares.

[Step Diagram]

Step 8: Glue up

Glue slices into the final panel, aligning the squares at every joint. Alternate clamps above and below the panel and check for flat with a straightedge.

Settings:
Clamp spacing: about 8" (200 mm)

NOTE: Use a waterproof PVA glue rated for indirect food contact. Clamp until squeeze-out turns rubbery, then scrape.

[Step Diagram]

Step 9: Flatten and sand

Flatten with a router sled or drum sander, never the thickness planer. Sand 80, 120, 180 grit. Break edges (180 grit).

Settings:

Grits: 80 / 120 / 180

DANGER: Never run an end-grain glue-up through the thickness planer. It can blow apart. Flatten with a router sled or drum sander.

[Step Diagram]

Step 10: Edge profile

Rout the edge profile, default: 1/8" (3 mm) roundover. Optional finger grips: core box / round-nose bit 3/8–1/2" (9.5–12.7 mm).

Settings:

Bit: 1/8" (3 mm) roundover

WARNING: Climb cuts can grab. Feed the router against the bit rotation and take shallow passes.

[Step Diagram]

Step 11: Finish

Raise the grain with a damp cloth, sand 180 again, then flood with food-safe mineral oil and top with board butter.

NOTE: Use closed-grain, non-toxic hardwoods only. Avoid spalted, softwood or oily/sensitising exotics on cutting surfaces.